Kilian (2007) Full-extension: Bayesian VAR Model

This notebook extends the original data workflow by estimating a Bayesian VAR with NIW prior and Gibbs sampling with extended data (2008-2025).

Setup

In [1]:
```
from pathlib import Path
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from scipy.stats import invwishart
from statsmodels.tsa.stattools import adfuller, kpss
from statsmodels.tsa.api import VAR
from statsmodels.stats.diagnostic import acorr_ljungbox

plt.style.use('seaborn-v0_8-whitegrid')
np.set_printoptions(suppress=True, precision=4)

BASE_DIR = Path("./data/")
```

Load the Kilian VAR data

The CPI data of 2025 oct is NaN because the government shutdown in 2025, so we truncate the series before 2025 october

In [2]:
```
df_old = pd.read_excel(BASE_DIR / 'data_1973_2007.xlsx')

# Convert obs like 1973M02 -> monthly datetime index (1973-02-01)
df_old = df_old.rename(columns={'dprod': 'd_prod'})
df_old['date'] = pd.to_datetime(
    df_old['obs'].astype(str).str.replace('M', '-', regex=False) + '-0
    format='%Y-%m-%d'
)

df_old = (
    df_old[['date', 'd_prod', 'rea', 'rpo']]
    .set_index('date')
    .sort_index()
)
```

```
df_old.head()
```

Out[2]:

	d_prod	rea	rpo
date			
1973-02-01	11.877264	36.064235	-47.999534
1973-03-01	1.419150	41.527016	-48.286474
1973-04-01	1.177711	43.991250	-47.082500
1973-05-01	27.455136	48.105889	-43.857651
1973-06-01	-13.110436	52.033462	-41.571063

In [3]:

```
df_new = pd.read_csv(BASE_DIR / 'data_extended_2008_2025.csv',
                     parse_dates=['date'], index_col='date')
# Drop Oct/Nov/Dec 2025 so the last rpo NaN (Oct 2025) and trailing mo
df_new = df_new[df_new.index < '2025-10-01']   # 2008-01 -> 2025-09

cols = ['d_prod', 'rea', 'rpo']

# Append old (1973-2007) and new (2008-2025) datasets
# keep='last' ensures newer data wins if there is any overlap
df = (
    pd.concat([df_old[cols], df_new[cols]], axis=0)
    .sort_index()
)
df = df[~df.index.duplicated(keep='last')]

df.head(),df.tail(),df.shape
```

Out[3]:

```
(               d_prod        rea        rpo
 date
 1973-02-01  11.877264  36.064235 -47.999534
 1973-03-01   1.419150  41.527016 -48.286474
 1973-04-01   1.177711  43.991250 -47.082500
 1973-05-01  27.455136  48.105889 -43.857651
 1973-06-01 -13.110436  52.033462 -41.571063,
                d_prod        rea         rpo
 date
 2025-05-01   1.291935 -25.227319 -164.619146
 2025-06-01  15.588897  -2.494234 -156.289244
 2025-07-01   6.183759   5.037654 -154.984964
 2025-08-01   6.464529  14.343198 -159.665539
 2025-09-01  15.695441  20.231109 -161.343980,
 (632, 3))
```

Data Summary and diagnostics

In [4]:

```
print(df.describe())
```

```
print(df.isnull().sum())
print(df.isna().sum())
```

```
           d_prod         rea         rpo
count  632.000000  632.000000  632.000000
mean     0.859636   -3.396694  -44.254331
std     18.973383   40.728404   75.822428
min   -173.885832 -161.243650 -272.750137
25%     -6.239976  -21.907556 -100.219742
50%      2.544631   -7.113553  -36.679731
75%      9.227303   14.396359   26.662418
max     77.983205  189.479230   90.636602
d_prod    0
rea       0
rpo       0
dtype: int64
d_prod    0
rea       0
rpo       0
dtype: int64
```

In [5]:

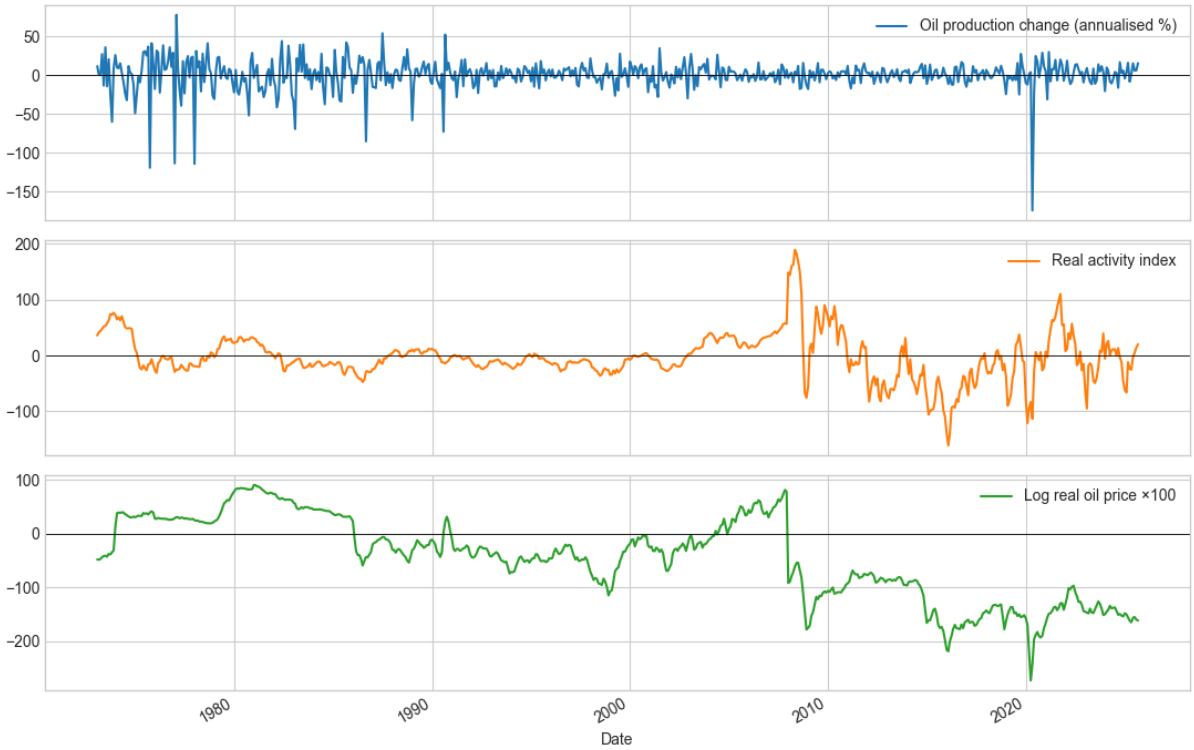
```
fig, axes = plt.subplots(3, 1, figsize=(11, 7), sharex=True)

axes[0].plot(df.index, df['d_prod'], label='Oil production change (ann
axes[0].axhline(0, color='k', lw=0.6)
axes[0].legend()

axes[1].plot(df.index, df['rea'], label='Real activity index', color='
axes[1].axhline(0, color='k', lw=0.6)
axes[1].legend()

axes[2].plot(df.index, df['rpo'], label='Log real oil price ×100', col
axes[2].axhline(0, color='k', lw=0.6)
axes[2].legend()

axes[2].set_xlabel('Date')
fig.autofmt_xdate()
plt.tight_layout()
plt.show()
```



Stationarity checks (ADF and KPSS)

In [6]:
```
rows = []
for c in cols:
    s = df[c].dropna().values
    adf = adfuller(s, regression='c', autolag='AIC')
    kps = kpss(s, regression='c', nlags='auto')
    rows.append({
        'series': c,
        'adf_stat': adf[0],
        'adf_pvalue': adf[1],
        'kpss_stat': kps[0],
        'kpss_pvalue': kps[1]
    })
stationarity_table = pd.DataFrame(rows)
stationarity_table
```

```
/var/folders/3g/9jy95tfs3s11qr_9lw9l2dsm0000gn/T/ipykernel_7382/9221278
69.py:5: InterpolationWarning: The test statistic is outside of the ran
ge of p-values available in the
look-up table. The actual p-value is greater than the p-value returned.

  kps = kpss(s, regression='c', nlags='auto')
/var/folders/3g/9jy95tfs3s11qr_9lw9l2dsm0000gn/T/ipykernel_7382/9221278
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/var/folders/3g/9jy95tfs3s11qr_9lw9l2dsm0000gn/T/ipykernel_7382/9221278
69.py:5: InterpolationWarning: The test statistic is outside of the ran
ge of p-values available in the
look-up table. The actual p-value is smaller than the p-value returned.

  kps = kpss(s, regression='c', nlags='auto')
```

Out[6]:

	series	adf_stat	adf_pvalue	kpss_stat	kpss_pvalue
0	d_prod	-6.026682	1.450329e-07	0.036947	0.10
1	rea	-4.417180	2.767273e-04	0.204174	0.10
2	rpo	-1.505072	5.310006e-01	2.619633	0.01

AIC table by lag length

In [7]:

```
aic_rows = []
for lag in range(1, 37):
    try:
        res = VAR(df[cols]).fit(lag)
        aic_rows.append({'lag': lag, 'AIC': res.aic})
    except Exception:
        aic_rows.append({'lag': lag, 'AIC': np.nan})
aic_table = pd.DataFrame(aic_rows)
print("Lag with minimum AIC:", aic_table.loc[aic_table['AIC'].idxmin()
aic_table
```

```
/Library/Frameworks/Python.framework/Versions/3.12/lib/python3.12/site-
packages/statsmodels/tsa/base/tsa_model.py:473: ValueWarning: No freque
ncy information was provided, so inferred frequency MS will be used.
  self._init_dates(dates, freq)
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packages/statsmodels/tsa/base/tsa_model.py:473: ValueWarning: No freque
ncy information was provided, so inferred frequency MS will be used.
  self._init_dates(dates, freq)
```

```
Lag with minimum AIC: 2
```

```
/Library/Frameworks/Python.framework/Versions/3.12/lib/python3.12/site-
packages/statsmodels/tsa/base/tsa_model.py:473: ValueWarning: No freque
ncy information was provided, so inferred frequency MS will be used.
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/Library/Frameworks/Python.framework/Versions/3.12/lib/python3.12/site-
packages/statsmodels/tsa/base/tsa_model.py:473: ValueWarning: No freque
ncy information was provided, so inferred frequency MS will be used.
  self._init_dates(dates, freq)
```

Out[7]:

	lag	AIC
0	1	16.098686
1	2	15.947494
2	3	15.950050
3	4	15.964984
4	5	15.955090
5	6	15.975818

6	7	15.985813
7	8	16.000328
8	9	15.986551
9	10	15.968143
10	11	15.981789
11	12	15.948540
12	13	15.960102
13	14	15.979509
14	15	15.970841
15	16	15.984601
16	17	16.000762
17	18	16.023135
18	19	16.043425
19	20	16.063904
20	21	16.082811
21	22	16.097283
22	23	16.114937
23	24	16.120712
24	25	16.096279
25	26	16.109142
26	27	16.132560
27	28	16.157781
28	29	16.169298
29	30	16.193631
30	31	16.214059
31	32	16.206200
32	33	16.167337
33	34	16.178998
34	35	16.197996
35	36	16.213181

BVAR(24) with NIW prior and Gibbs sampling

In [8]:
```
def make_var_regressors(y, p):
    T, n = y.shape
    Y = y[p:, :]
    X_lags = []
    for t in range(p, T):
        row = []
        for L in range(1, p + 1):
            row.extend(y[t - L, :])
        X_lags.append(row)
    X_lags = np.asarray(X_lags)
    X = np.column_stack([np.ones(X_lags.shape[0]), X_lags])
    return Y, X

def draw_matrix_normal(M, V, Sigma, rng):
    k, n = M.shape
    Lv = np.linalg.cholesky(V)
    Ls = np.linalg.cholesky(Sigma)
    Z = rng.standard_normal((k, n))
    return M + Lv @ Z @ Ls.T

def gibbs_bvar_niw(y, p=24, n_draws=12000, burn=2000, seed=42):
    rng = np.random.default_rng(seed)
    Y, X = make_var_regressors(y, p)
    T_eff, n = Y.shape
    k = X.shape[1]

    # NIW prior hyperparameters
    B0 = np.zeros((k, n)) # starting point of the draw is 0
    V0 = np.eye(k) * 10.0 # variance of the prior is 10 %*% I
    V0_inv = np.linalg.inv(V0)
    nu0 = n + 2 # degrees of freedom of the prior
    S0 = np.eye(n) # scale matrix of the prior

    # OLS initialization
    B_ols = np.linalg.inv(X.T @ X) @ (X.T @ Y)
    E_ols = Y - X @ B_ols
    Sigma = (E_ols.T @ E_ols) / T_eff
    B = B_ols.copy()

    keep = n_draws - burn
    B_draws = np.zeros((keep, k, n))
    Sigma_draws = np.zeros((keep, n, n))
    B_trace = np.zeros((n_draws, 4))
    S_trace = np.zeros((n_draws, n))

    XtX = X.T @ X
    XtY = X.T @ Y

    for it in range(n_draws):
```

```
        Vn = np.linalg.inv(V0_inv + XtX)
        Bn = Vn @ (V0_inv @ B0 + XtY)
        B = draw_matrix_normal(Bn, Vn, Sigma, rng)

        E = Y - X @ B
        diff = B - B0
        Sn = S0 + E.T @ E + diff.T @ V0_inv @ diff
        nun = nu0 + T_eff + k
        Sigma = invwishart.rvs(df=nun, scale=Sn, random_state=rng)

        B_trace[it, :] = [B[0,0], B[1,0], B[1,1], B[2,2]]
        S_trace[it, :] = np.diag(Sigma)

        if it >= burn:
            j = it - burn
            B_draws[j] = B
            Sigma_draws[j] = Sigma

    out = {
        'Y': Y, 'X': X,
        'B_draws': B_draws,
        'Sigma_draws': Sigma_draws,
        'B_mean': B_draws.mean(axis=0),
        'Sigma_mean': Sigma_draws.mean(axis=0),
        'B_trace': B_trace,
        'S_trace': S_trace
    }
    return out

p = 24
n_draws = 12000
burn = 2000

# df[cols].values produces a pure float64 array — required for matrix
bvar = gibbs_bvar_niw(df[cols].values.astype(float), p=p, n_draws=n_dr
bvar['B_mean'].shape, bvar['Sigma_mean']
```

Out[8]:

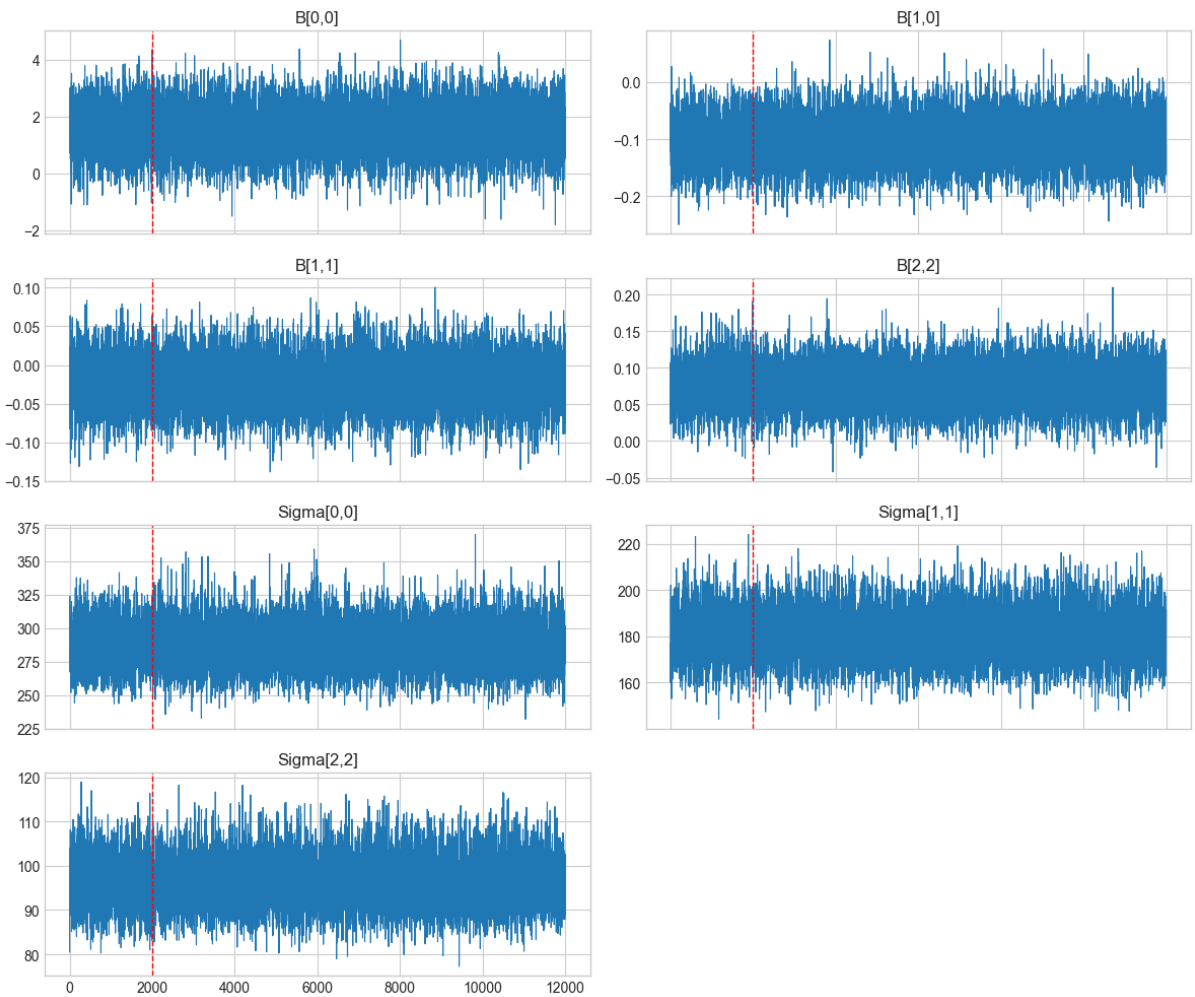
```
((73, 3),
 array([[286.7751,   7.9039, -17.3809],
        [  7.9039, 180.5206, -16.0293],
        [-17.3809, -16.0293,  96.3122]]))
```

MCMC trace plots (visual convergence check)

In [9]:

```
fig, axes = plt.subplots(4, 2, figsize=(12, 10), sharex=True)
axes = axes.ravel()

trace_names = ['B[0,0]', 'B[1,0]', 'B[1,1]', 'B[2,2]']
for i in range(4):
    axes[i].plot(bvar['B_trace'][:, i], lw=0.8)
    axes[i].axvline(2000, color='r', ls='--', lw=1)
```

```
    axes[i].set_title(trace_names[i])

sig_names = ['Sigma[0,0]', 'Sigma[1,1]', 'Sigma[2,2]']
for i in range(3):
    axes[4+i].plot(bvar['S_trace'][:, i], lw=0.8)
    axes[4+i].axvline(2000, color='r', ls='--', lw=1)
    axes[4+i].set_title(sig_names[i])

axes[-1].axis('off')
plt.tight_layout()
plt.show()
```



Residual diagnostics (Ljung-Box test)

In [10]:
```
Y = bvar['Y']
X = bvar['X']
Bhat = bvar['B_mean']
resid = Y - X @ Bhat

lb_rows = []
for i, c in enumerate(cols):
    lb = acorr_ljungbox(resid[:, i], lags=[12], return_df=True)
    lb_rows.append({
```

```
        'series': c,
        'lb_stat_lag12': float(lb['lb_stat'].iloc[0]),
        'lb_pvalue_lag12': float(lb['lb_pvalue'].iloc[0])
    })
ljung_box_table = pd.DataFrame(lb_rows)
ljung_box_table
```

Out[10]:

	series	lb_stat_lag12	lb_pvalue_lag12
0	d_prod	2.328304	0.998711
1	rea	0.832569	0.999995
2	rpo	0.118393	1.000000

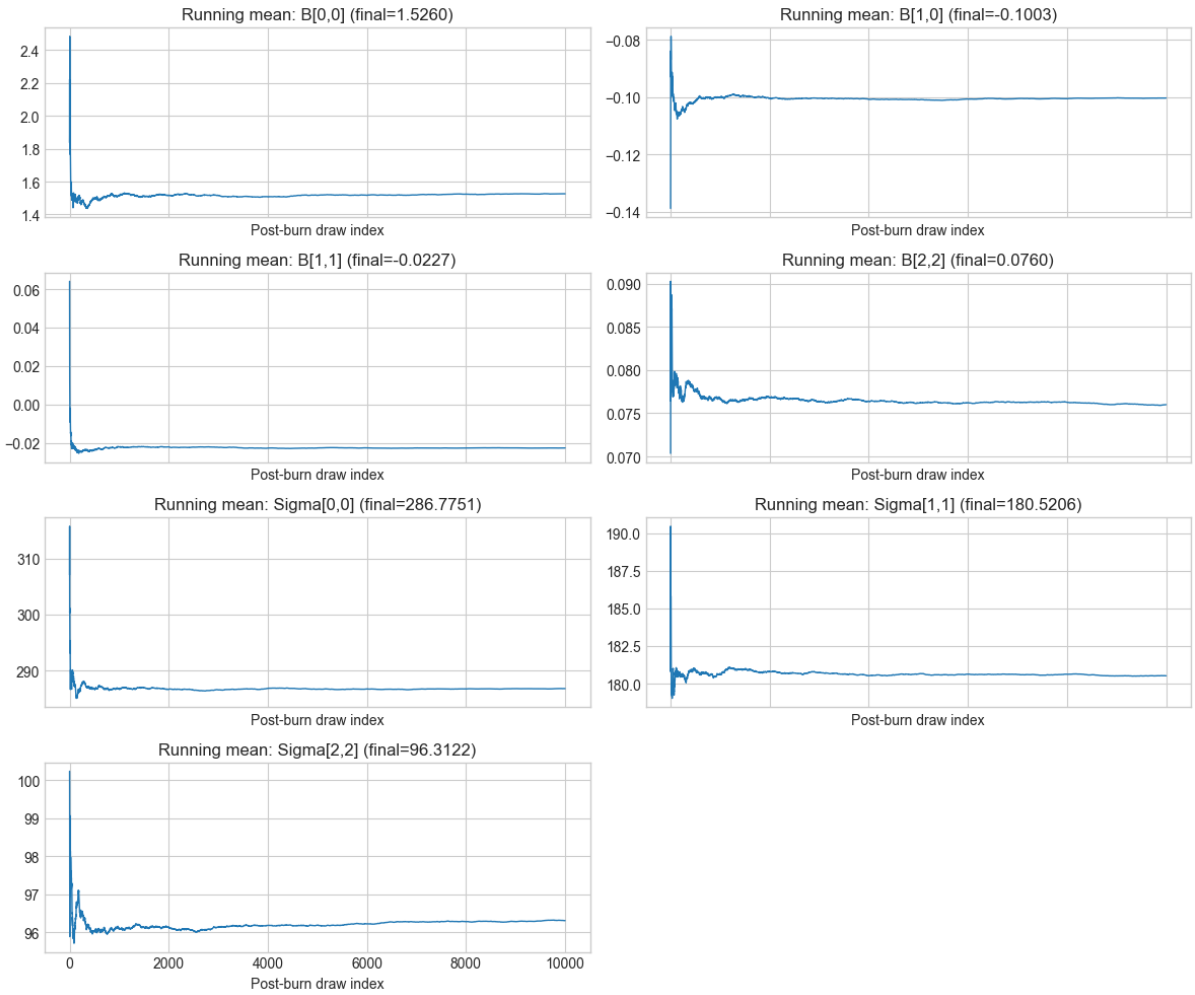
Running posterior mean over iterations

This plots the cumulative posterior mean (after burn-in) to visually inspect stabilization over MCMC draws.

In [11]:

```
keep_B = bvar['B_draws']
keep_S = bvar['Sigma_draws']

cum_mean_B00 = np.cumsum(keep_B[:, 0, 0]) / np.arange(1, keep_B.shape[
cum_mean_B10 = np.cumsum(keep_B[:, 1, 0]) / np.arange(1, keep_B.shape[
cum_mean_B11 = np.cumsum(keep_B[:, 1, 1]) / np.arange(1, keep_B.shape[
cum_mean_B22 = np.cumsum(keep_B[:, 2, 2]) / np.arange(1, keep_B.shape[

cum_mean_S00 = np.cumsum(keep_S[:, 0, 0]) / np.arange(1, keep_S.shape[
cum_mean_S11 = np.cumsum(keep_S[:, 1, 1]) / np.arange(1, keep_S.shape[
cum_mean_S22 = np.cumsum(keep_S[:, 2, 2]) / np.arange(1, keep_S.shape[

fig, axes = plt.subplots(4, 2, figsize=(12, 10), sharex=True)
axes = axes.ravel()

series = [
    (cum_mean_B00, 'Running mean: B[0,0]'),
    (cum_mean_B10, 'Running mean: B[1,0]'),
    (cum_mean_B11, 'Running mean: B[1,1]'),
    (cum_mean_B22, 'Running mean: B[2,2]'),
    (cum_mean_S00, 'Running mean: Sigma[0,0]'),
    (cum_mean_S11, 'Running mean: Sigma[1,1]'),
    (cum_mean_S22, 'Running mean: Sigma[2,2]'),
]

for i, (s, ttl) in enumerate(series):
    axes[i].plot(s, lw=1)
    axes[i].set_title(f"{ttl} (final={s[-1]:.4f})")
    axes[i].set_xlabel('Post-burn draw index')
```

```
axes[-1].axis('off')
plt.tight_layout()
plt.show()
```



In [12]:
```
def bvar_state_matrices(B_mat, n=3, p=24):
    """Convert B matrix (k x n) into intercept and companion A."""
    c = B_mat[0, :]  # intercept (n,)
    A_top = B_mat[1:, :].T  # (n, n*p)
    if p == 1:
        A_comp = A_top
    else:
        bottom = np.hstack([np.eye(n * (p - 1)), np.zeros((n * (p - 1)
        A_comp = np.vstack([A_top, bottom])
    return c, A_top, A_comp


def irf_from_companion(A_comp, Sigma, n=3, p=24, h=15):
    """Cholesky-identified IRFs as (n^2, h+1) stacked by columns."""
    J = np.hstack([np.eye(n), np.zeros((n, n * p - n))])
    C = np.linalg.cholesky(Sigma).T
    comp_power = np.eye(n * p)
    irfs = []
    for _ in range(h + 1):
        resp = J @ comp_power @ J.T @ C
```

```
        irfs.append(resp.reshape(n * n, 1, order='F'))
        comp_power = comp_power @ A_comp
    return np.hstack(irfs)


def make_var_xy(y, p):
    """Build Y, X with X=[1, y_{t-1},...,y_{t-p}] consistent with note
    T, n = y.shape
    Y = y[p:, :]
    X_lags = []
    for t in range(p, T):
        row = []
        for L in range(1, p + 1):
            row.extend(y[t - L, :])
        X_lags.append(row)
    X = np.column_stack([np.ones(len(X_lags)), np.asarray(X_lags)])
    return Y, X


def structural_shocks_from_bmean(y, B_mean, Sigma_mean, p=24):
    Y, X = make_var_xy(y, p)
    U = (Y - X @ B_mean).T  # n x (T-p)
    C = np.linalg.cholesky(Sigma_mean).T
    Ehat = np.linalg.solve(C, U)
    return Ehat


def stage2irf(y, q_shock, nrep=20000, seed=42):
    y = np.asarray(y).reshape(-1, 1)
    q_shock = np.asarray(q_shock).reshape(-1, 1)
    t = len(q_shock)
    pp = 12

    Z = [np.ones((t - pp, 1))]
    for i in range(pp + 1):
        Z.append(q_shock[pp - i:t - i, :])
    Z = np.hstack(Z)

    y_dep = y[pp:t, :]
    bhat = np.linalg.inv(Z.T @ Z) @ Z.T @ y_dep

    irf1hat = bhat[1:, 0]
    cumirf1hat = np.cumsum(irf1hat)

    rng = np.random.default_rng(seed)
    IRF1 = np.zeros((nrep, 13))
    cumIRF1 = np.zeros((nrep, 13))
    block = 4
    y_len = len(y_dep)

    for j in range(nrep):
        yr_blocks, Zr_blocks = [], []
```

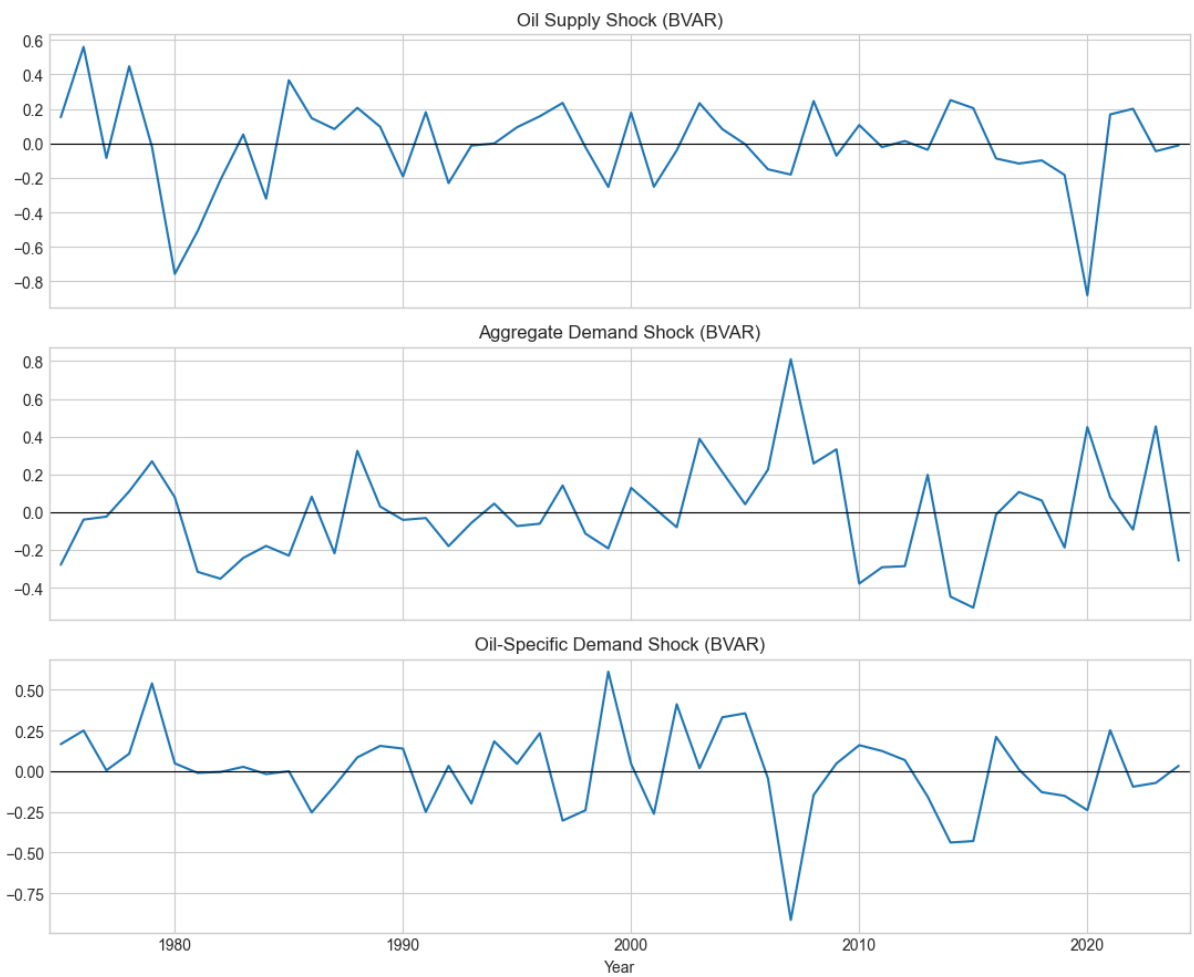
```
        n_blocks = int(np.ceil(y_len / block))
        for _ in range(n_blocks):
            pos = int(np.ceil(rng.random() * (y_len - block)))
            start = max(0, min(pos - 1, y_len - block))
            yr_blocks.append(y_dep[start:start + block, :])
            Zr_blocks.append(Z[start:start + block, :])

        yr = np.vstack(yr_blocks)[:y_len, :]
        Zr = np.vstack(Zr_blocks)[:y_len, :]

        br = np.linalg.inv(Zr.T @ Zr) @ Zr.T @ yr
        IRF1[j, :] = br[1:, 0]
        cumIRF1[j, :] = np.cumsum(IRF1[j, :])

    q16, q84 = np.quantile(IRF1, [0.16, 0.84], axis=0)
    q025, q975 = np.quantile(IRF1, [0.025, 0.975], axis=0)
    c16, c84 = np.quantile(cumIRF1, [0.16, 0.84], axis=0)
    c025, c975 = np.quantile(cumIRF1, [0.025, 0.975], axis=0)

    return {
        'irf': irf1hat,
        'irf_68': np.vstack([q16, q84]),
        'irf_95': np.vstack([q025, q975]),
        'cum': cumirf1hat,
        'cum_68': np.vstack([c16, c84]),
        'cum_95': np.vstack([c025, c975]),
    }
```

Bayesian Figure 2: structural shocks (posterior mean parameters)

In [13]:

```
p = 24
n = 3
h = 15

# Use only numeric columns — df[cols].values is float64
y = df[cols].values.astype(float)
Ehat_b = structural_shocks_from_bmean(y, bvar['B_mean'], bvar['Sigma_m

# Date axis: shocks start after p lags; df.index[p] is the first obser
shock_dates = df.index[p:]

# Annual averages of monthly structural shocks
# We can only use complete 12-month blocks
n_shocks = Ehat_b.shape[1]  # number of shock observations available
n_full_years = n_shocks // 12
n_months_to_use = n_full_years * 12

year_min = shock_dates[0].year
time_year = np.arange(year_min, year_min + n_full_years)
```

```
def annual_avg(shock_row, n_months):
    """Average monthly shocks over complete years."""
    return shock_row[:n_months].reshape(-1, 12).mean(axis=1)

q1a = annual_avg(Ehat_b[0], n_months_to_use)
q2a = annual_avg(Ehat_b[1], n_months_to_use)
q3a = annual_avg(Ehat_b[2], n_months_to_use)

fig, axes = plt.subplots(3, 1, figsize=(11, 9), sharex=True)
for ax, s, ttl in zip(
    axes,
    [q1a, q2a, q3a],
    ['Oil Supply Shock (BVAR)', 'Aggregate Demand Shock (BVAR)', 'Oil-
):
    ax.plot(time_year, s) # use line plot instead
    ax.axhline(0, color='k', lw=0.8)
    ax.set_xlim(year_min - 0.5, time_year[-1] + 0.5)
    ax.set_title(ttl)

axes[-1].set_xlabel('Year')
plt.tight_layout()
plt.show()

print(f"Showing annual averages for {n_full_years} complete years ({ye
```

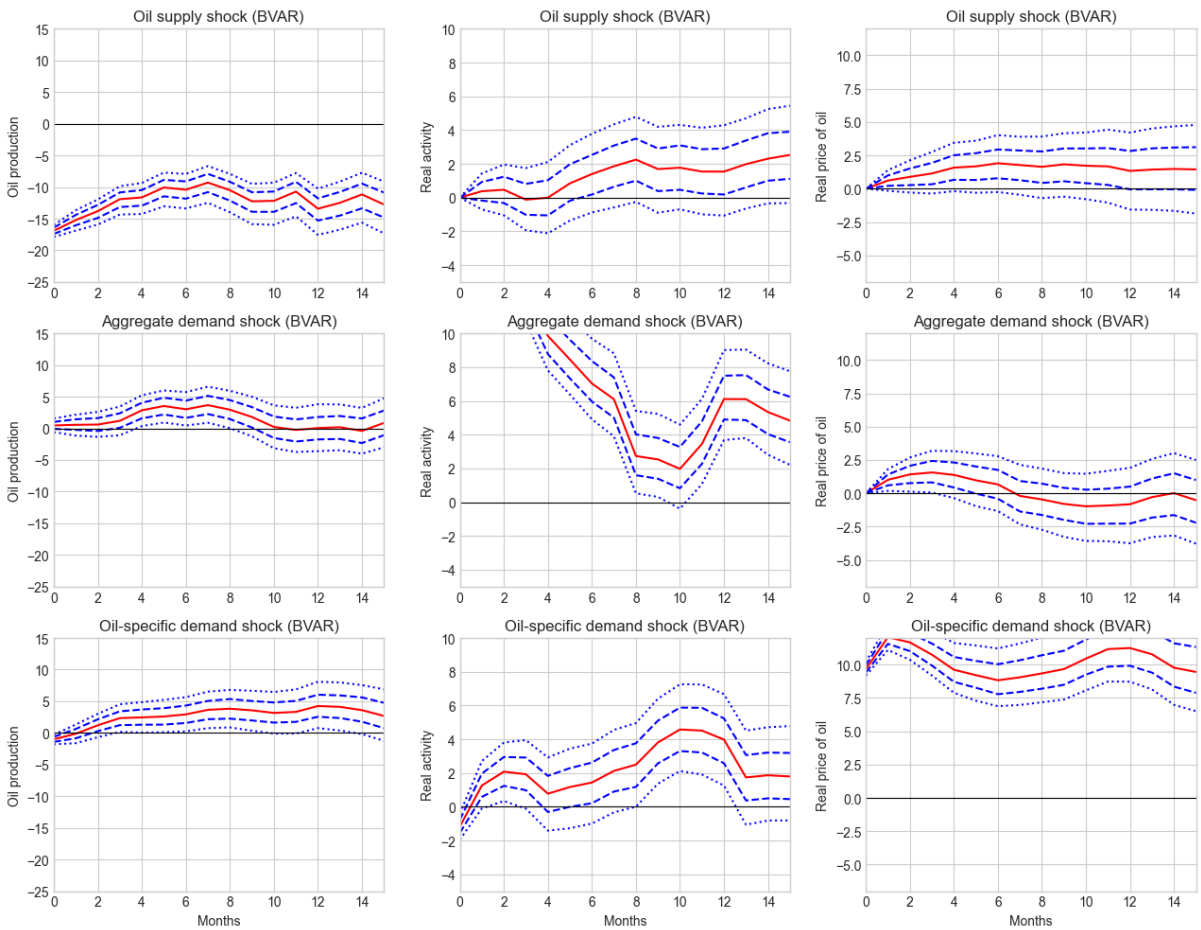


```
Showing annual averages for 50 complete years (1975–2024)
```

Bayesian Figure 3: IRFs and credible intervals from posterior draws

In [14]:
```
# Posterior draw IRFs for credible bands
n_keep = bvar['B_draws'].shape[0]
max_draws = min(2000, n_keep)
idx = np.linspace(0, n_keep - 1, max_draws, dtype=int)

IRF_draws = np.zeros((max_draws, n * n, h + 1))
for j, d in enumerate(idx):
    B_d = bvar['B_draws'][d]
    S_d = bvar['Sigma_draws'][d]
    _, _, A_comp_d = bvar_state_matrices(B_d, n=n, p=p)
    irf_d = irf_from_companion(A_comp_d, S_d, n=n, p=p, h=h)
    irf_d[0, :] = np.cumsum(irf_d[0, :])
    irf_d[3, :] = np.cumsum(irf_d[3, :])
    irf_d[6, :] = np.cumsum(irf_d[6, :])
    IRF_draws[j] = irf_d

IRF_med = np.median(IRF_draws, axis=0)
IRF_68 = np.quantile(IRF_draws, [0.16, 0.84], axis=0)
IRF_95 = np.quantile(IRF_draws, [0.025, 0.975], axis=0)

horizon = np.arange(0, h + 1)
fig, axes = plt.subplots(3, 3, figsize=(13, 10))
axes = axes.ravel()

meta = [
    ('Oil supply shock (BVAR)', 'Oil production', (-25, 15), -1),
    ('Oil supply shock (BVAR)', 'Real activity', (-5, 10), -1),
    ('Oil supply shock (BVAR)', 'Real price of oil', (-7, 12), -1),
    ('Aggregate demand shock (BVAR)', 'Oil production', (-25, 15), 1),
    ('Aggregate demand shock (BVAR)', 'Real activity', (-5, 10), 1),
    ('Aggregate demand shock (BVAR)', 'Real price of oil', (-7, 12), 1
    ('Oil-specific demand shock (BVAR)', 'Oil production', (-25, 15),
    ('Oil-specific demand shock (BVAR)', 'Real activity', (-5, 10), 1)
    ('Oil-specific demand shock (BVAR)', 'Real price of oil', (-7, 12)
]

for i, (ttl, yl, ylim, sgn) in enumerate(meta):
    ax = axes[i]
    m = sgn * IRF_med[i]
    l68, u68 = sgn * IRF_68[0, i], sgn * IRF_68[1, i]
    l95, u95 = sgn * IRF_95[0, i], sgn * IRF_95[1, i]
    lo68, hi68 = np.minimum(l68, u68), np.maximum(l68, u68)
    lo95, hi95 = np.minimum(l95, u95), np.maximum(l95, u95)

    ax.plot(horizon, m, 'r-')
    ax.plot(horizon, lo68, 'b--')
```

```
    ax.plot(horizon, hi68, 'b--')
    ax.plot(horizon, lo95, 'b:')
    ax.plot(horizon, hi95, 'b:')
    ax.axhline(0, color='k', lw=0.8)
    ax.set_title(ttl)
    ax.set_ylabel(yl)
    ax.set_xlim(0, h)
    ax.set_ylim(*ylim)
    if i >= 6:
        ax.set_xlabel('Months')

plt.tight_layout()
plt.show()
```

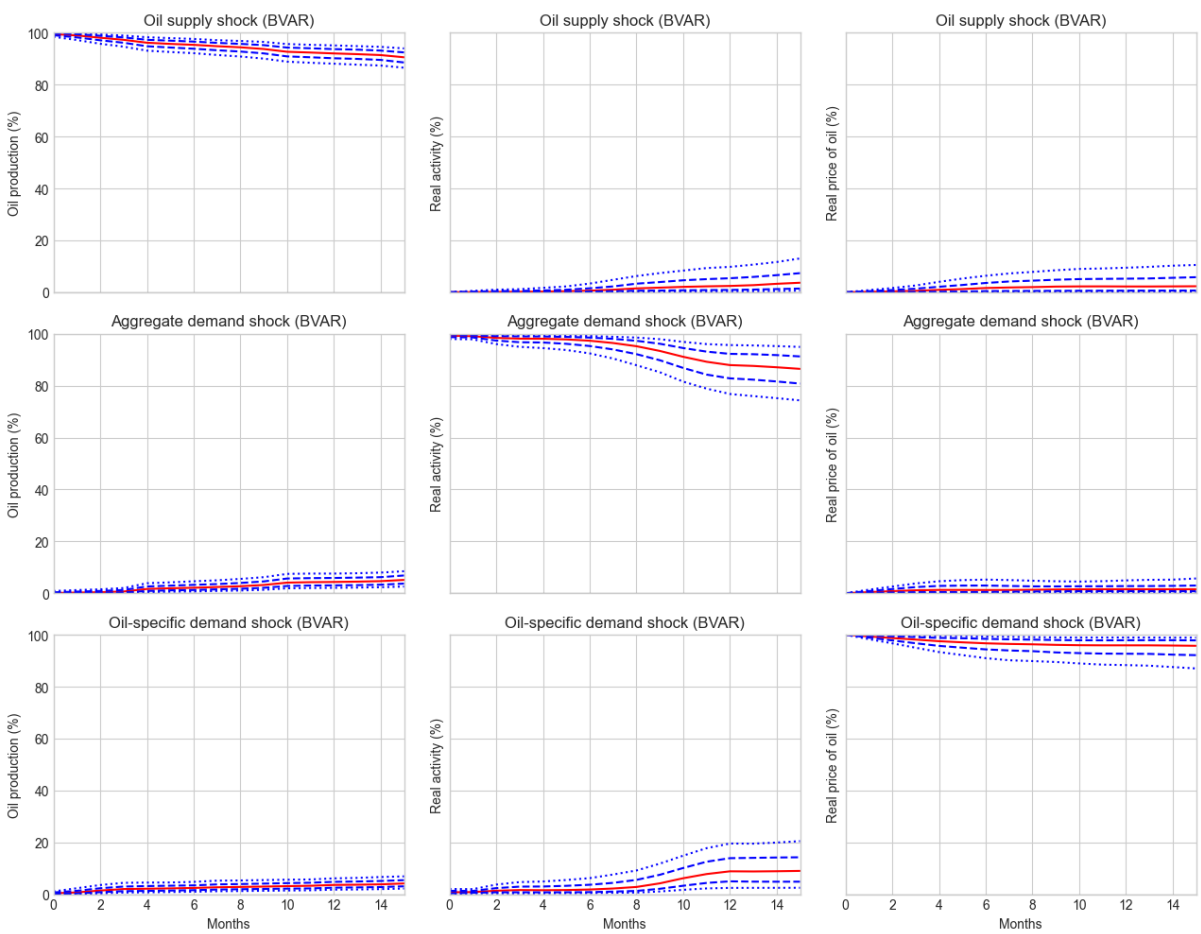


4.1 FEVD

In [15]:
```
# FEVD: posterior draws + credible bands
n_keep = bvar['B_draws'].shape[0]
max_draws = min(2000, n_keep)
idx = np.linspace(0, n_keep - 1, max_draws, dtype=int)

# FEVD_draws[d, shock, variable, horizon]
FEVD_draws = np.zeros((max_draws, n, n, h + 1))
```

```
for j, d in enumerate(idx):
    B_d = bvar['B_draws'][d]
    S_d = bvar['Sigma_draws'][d]
    _, _, A_comp_d = bvar_state_matrices(B_d, n=n, p=p)
    irf_d = irf_from_companion(A_comp_d, S_d, n=n, p=p, h=h)

    # irf_d uses order='F': [var1_shock1, var2_shock1, var3_shock1, va
    # Reshape to (variable, shock, horizon) then transpose to (shock,
    psi = irf_d.reshape(n, n, h + 1, order='F').transpose(1, 0, 2)

    # cumulative squared impulse responses up to horizon t
    cum_sq = np.cumsum(psi ** 2, axis=2)
    denom = np.sum(cum_sq, axis=0, keepdims=True)

    # avoid divide-by-zero at very early horizons
    FEVD_draws[j] = np.divide(cum_sq, denom, out=np.zeros_like(cum_sq)

FEVD_med = np.median(FEVD_draws, axis=0)
FEVD_68 = np.quantile(FEVD_draws, [0.16, 0.84], axis=0)
FEVD_95 = np.quantile(FEVD_draws, [0.025, 0.975], axis=0)

horizon = np.arange(0, h + 1)
fig, axes = plt.subplots(3, 3, figsize=(13, 10), sharex=True, sharey=T
axes = axes.ravel()

meta = [
    ('Oil supply shock (BVAR)', 'Oil production'),
    ('Oil supply shock (BVAR)', 'Real activity'),
    ('Oil supply shock (BVAR)', 'Real price of oil'),
    ('Aggregate demand shock (BVAR)', 'Oil production'),
    ('Aggregate demand shock (BVAR)', 'Real activity'),
    ('Aggregate demand shock (BVAR)', 'Real price of oil'),
    ('Oil-specific demand shock (BVAR)', 'Oil production'),
    ('Oil-specific demand shock (BVAR)', 'Real activity'),
    ('Oil-specific demand shock (BVAR)', 'Real price of oil'),
]

for k, (ttl, yl) in enumerate(meta):
    shock = k // n
    var = k % n

    med = 100.0 * FEVD_med[shock, var]
    l68 = 100.0 * FEVD_68[0, shock, var]
    u68 = 100.0 * FEVD_68[1, shock, var]
    l95 = 100.0 * FEVD_95[0, shock, var]
    u95 = 100.0 * FEVD_95[1, shock, var]

    lo68, hi68 = np.minimum(l68, u68), np.maximum(l68, u68)
    lo95, hi95 = np.minimum(l95, u95), np.maximum(l95, u95)

    ax = axes[k]
    ax.plot(horizon, med, 'r-')
```

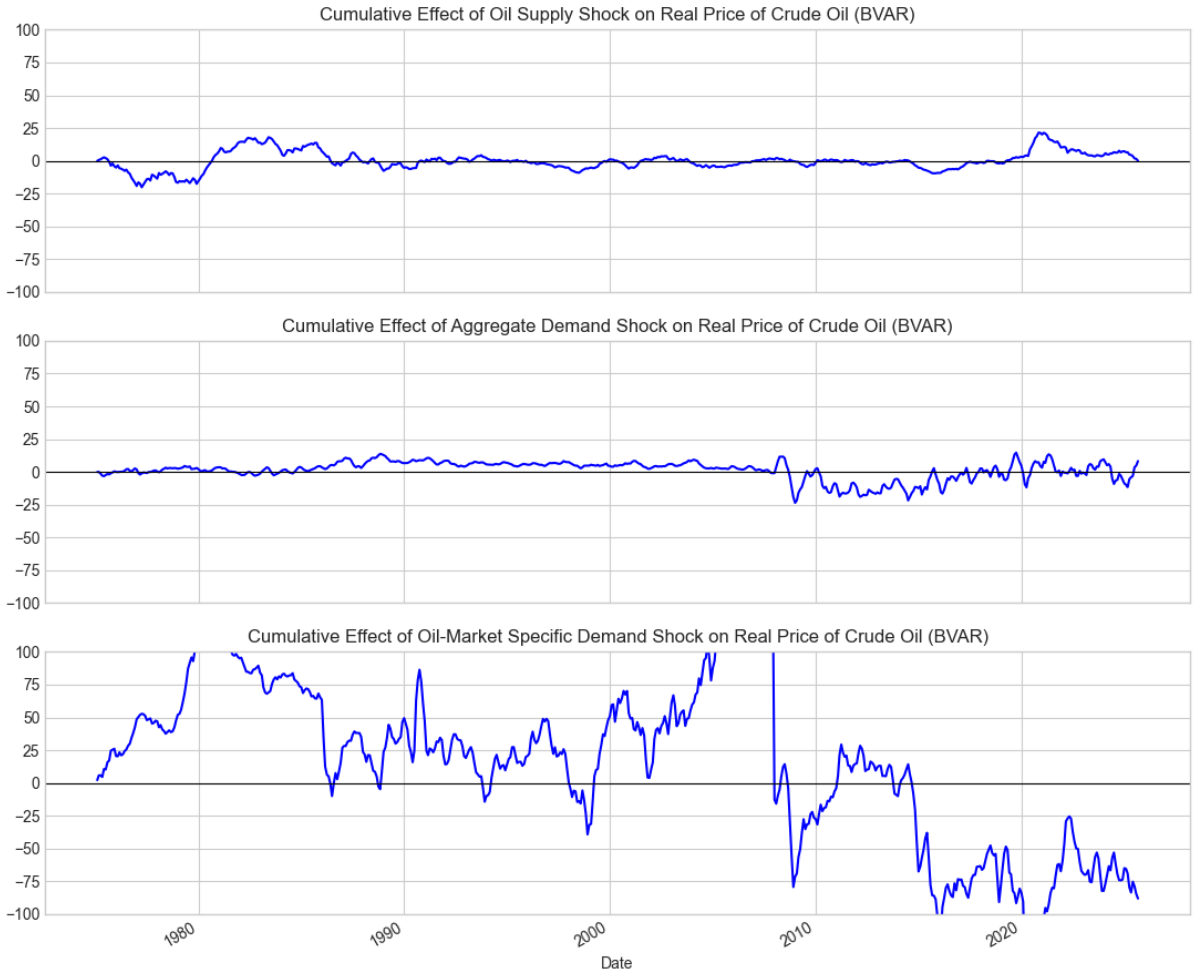
```
    ax.plot(horizon, lo68, 'b--')
    ax.plot(horizon, hi68, 'b--')
    ax.plot(horizon, lo95, 'b:')
    ax.plot(horizon, hi95, 'b:')
    ax.axhline(0, color='k', lw=0.8)
    ax.set_title(ttl)
    ax.set_ylabel(f"{yl} (%)")
    ax.set_xlim(0, h)
    ax.set_ylim(0, 100)
    if k >= 6:
        ax.set_xlabel('Months')

plt.tight_layout()
plt.show()

# Verification: FEVD should sum to 100% across shocks for each variabl
print("FEVD verification (should sum to ~100% for each variable at eac
for var in range(n):
    var_names = ['Oil production', 'Real activity', 'Real price of oil
    fevd_sum = np.sum(FEVD_med[:, var, :], axis=0) * 100
    print(f"{var_names[var]}: h=0: {fevd_sum[0]:.1f}%, h=6: {fevd_sum[
```




```
FEVD verification (should sum to ~100% for each variable at each horizo
n):
Oil production: h=0: 99.9%, h=6: 99.9%, h=15: 99.8%
Real activity: h=0: 100.0%, h=6: 99.7%, h=15: 99.1%
Real price of oil: h=0: 100.0%, h=6: 99.6%, h=15: 99.6%
```

Bayesian Figure 4: cumulative effect on real oil price

In [16]:
```
T = y.shape[0]
_, _, A_comp_mean = bvar_state_matrices(bvar['B_mean'], n=n, p=p)
IRF_long = irf_from_companion(A_comp_mean, bvar['Sigma_mean'], n=n, p=

yhat1 = np.zeros(T - p)
yhat2 = np.zeros(T - p)
yhat3 = np.zeros(T - p)
for i in range(T - p):
    idx_i = np.arange(i + 1)
    yhat1[i] = np.dot(IRF_long[2, idx_i], Ehat_b[0, i::-1])
    yhat2[i] = np.dot(IRF_long[5, idx_i], Ehat_b[1, i::-1])
    yhat3[i] = np.dot(IRF_long[8, idx_i], Ehat_b[2, i::-1])

# Date axis aligned to the actual data: first shock date is df.index[p
time_dates = df.index[p:]

fig, axes = plt.subplots(3, 1, figsize=(11, 9), sharex=True)

titles = [
    'Cumulative Effect of Oil Supply Shock on Real Price of Crude Oil
    'Cumulative Effect of Aggregate Demand Shock on Real Price of Crud
    'Cumulative Effect of Oil-Market Specific Demand Shock on Real Pri
]
for ax, yhat, ttl in zip(axes, [yhat1, yhat2, yhat3], titles):
    ax.plot(time_dates, yhat, 'b-')
    ax.axhline(0, color='k', lw=0.8)
    ax.set_title(ttl)
    ax.set_ylim(-100, 100)

axes[-1].set_xlabel('Date')
fig.autofmt_xdate()
plt.tight_layout()
plt.show()
```



Bayesian Figure 5-style responses: shocks on real GDP and CPI

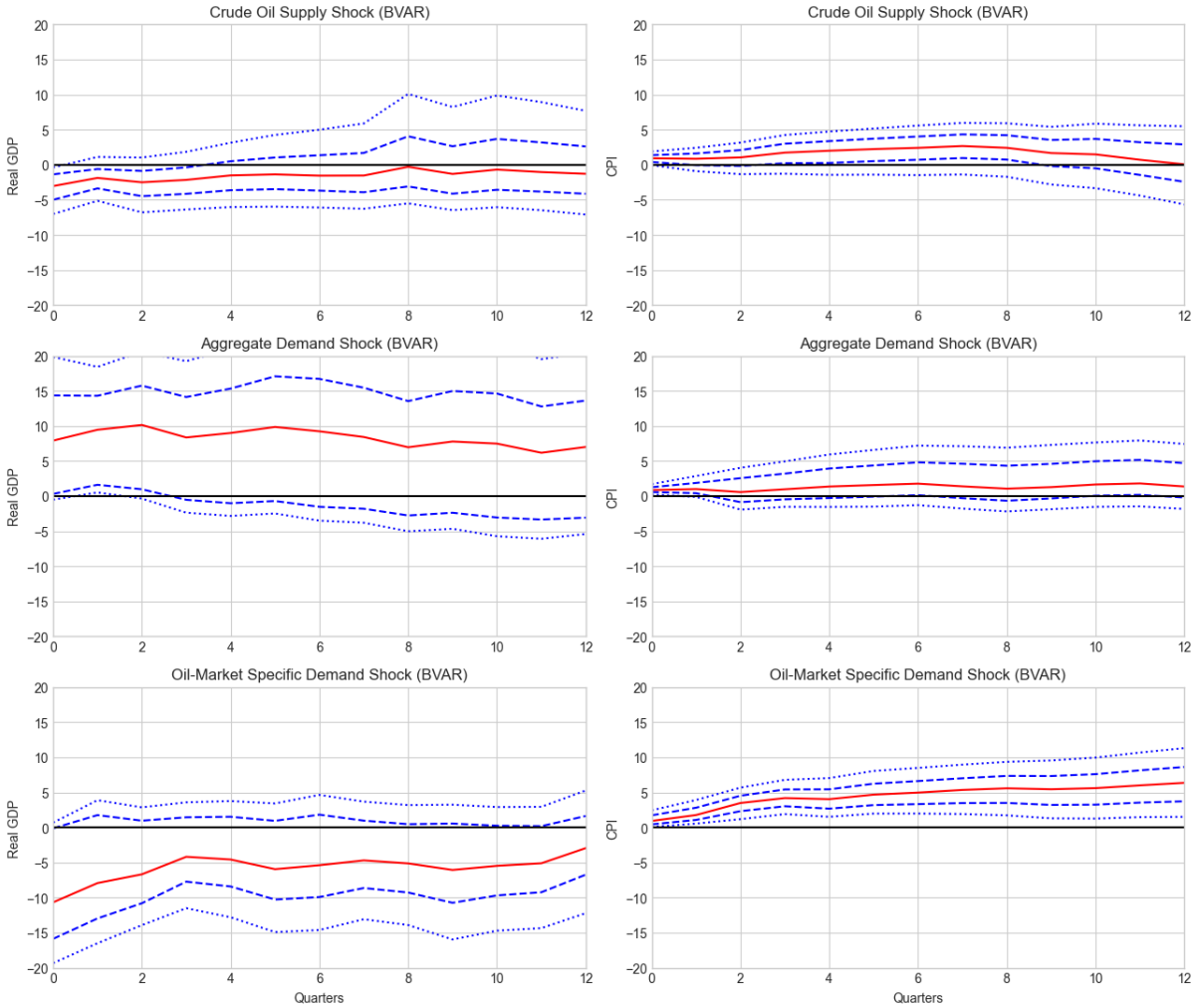
In [17]:
```
# Load merged GDP and CPI data (Kilian's original 1974–2007 + extensio
# These files were created by fetching GDPC1 and CPIAUCSL from FRED

# GDP: quarterly real GDP in billions of chained dollars
beagdp_merged = np.loadtxt(BASE_DIR / 'beagdp_merged_1974_2025.txt')
print("GDP data loaded:")
print(f"  Range: {int(beagdp_merged[0,0])} Q{int(beagdp_merged[0,1])}
      f"{int(beagdp_merged[-1,0])} Q{int(beagdp_merged[-1,1])}")
print(f"  Quarters: {len(beagdp_merged)}")

# CPI: monthly CPI-U (all urban consumers)
cpi_merged = np.loadtxt(BASE_DIR / 'cpi_merged_1974_2025.txt')
print("\nCPI data loaded:")
print(f"  Range: {int(cpi_merged[0,0])}-{int(cpi_merged[0,1]):02d} → "
      f"{int(cpi_merged[-1,0])}-{int(cpi_merged[-1,1]):02d}")
print(f"  Months: {len(cpi_merged)}")
```

```
GDP data loaded:
  Range: 1974 Q4 → 2025 Q4
  Quarters: 205

CPI data loaded:
  Range: 1974-12 → 2025-12
  Months: 612
```

In [18]:

```
# ═══════════════════════════════════════════════════════════════════
# Stage-2 setup: quarterly oil shocks → GDP growth and CPI inflation
# ═══════════════════════════════════════════════════════════════════

# Quarterly average structural shocks from monthly BVAR estimates
shock_dates = df.index[p:]  # first shock: 2010-01 (after p=24 lags fr

# Pad with initial value (Kilian convention: average first two months)
q1 = np.concatenate([[(Ehat_b[0, 0] + Ehat_b[0, 1]) / 2], Ehat_b[0, :]
q2 = np.concatenate([[(Ehat_b[1, 0] + Ehat_b[1, 1]) / 2], Ehat_b[1, :]
q3 = np.concatenate([[(Ehat_b[2, 0] + Ehat_b[2, 1]) / 2], Ehat_b[2, :]

# Quarterly averages (drop incomplete trailing months)
nq = len(q1) // 3
q1q = q1[:3 * nq].reshape(-1, 3).mean(axis=1)
q2q = q2[:3 * nq].reshape(-1, 3).mean(axis=1)
q3q = q3[:3 * nq].reshape(-1, 3).mean(axis=1)

# ───────────────────────────────────────────────────────────────────
# GDP growth rate (quarterly, annualized %)
# ───────────────────────────────────────────────────────────────────
gdp_vals = beagdp_merged[:, 2]  # column 3: real GDP
y_gdp_full = (np.log(gdp_vals[1:]) - np.log(gdp_vals[:-1])) * 400

# Align with quarterly shocks: first shock quarter is 2010 Q1
first_shock_year = shock_dates[0].year
first_shock_qtr = (shock_dates[0].month - 1) // 3 + 1

gdp_start_year = int(beagdp_merged[0, 0])
gdp_start_qtr = int(beagdp_merged[0, 1])

# Quarters from GDP start to first shock
offset_q = (first_shock_year - gdp_start_year) * 4 + (first_shock_qtr

# Extract aligned GDP growth (same length as quarterly shocks)
y_gdp = y_gdp_full[offset_q - 1 : offset_q - 1 + nq, None]

# ───────────────────────────────────────────────────────────────────
# CPI inflation rate (quarterly, annualized %)
# ───────────────────────────────────────────────────────────────────
cpi_vals = cpi_merged[:, 2]
cpi_log = np.log(cpi_vals)

# Quarterly inflation: (log(CPI_end) - log(CPI_start)) × 400 for each
first_cpi_idx = offset_q * 3  # months from CPI start to first shock q
```

```
infl_q = []
for i in range(nq):
    idx_start = first_cpi_idx + i * 3
    idx_end = idx_start + 3
    if idx_end < len(cpi_log) and not np.isnan(cpi_vals[idx_end]):
        infl_q.append((cpi_log[idx_end] - cpi_log[idx_start]) * 400)
    else:
        # Hit NaN or end of data — stop here
        break

y_cpi = np.array(infl_q)[:, None]

print(f"\nStage-2 alignment:")
print(f"  Quarterly shocks : {nq} quarters (q1q, q2q, q3q)")
print(f"  GDP growth       : {y_gdp.shape[0]} quarters")
print(f"  CPI inflation    : {y_cpi.shape[0]} quarters")
print(f"  First quarter    : {first_shock_year} Q{first_shock_qtr}")

# Align lengths explicitly to avoid mismatch errors in stage2irf
nq_gdp = min(len(q1q), y_gdp.shape[0])
nq_cpi = min(len(q1q), y_cpi.shape[0])

q1q_gdp, q2q_gdp, q3q_gdp = q1q[:nq_gdp], q2q[:nq_gdp], q3q[:nq_gdp]
q1q_cpi, q2q_cpi, q3q_cpi = q1q[:nq_cpi], q2q[:nq_cpi], q3q[:nq_cpi]

y_gdp_use = y_gdp[:nq_gdp]
y_cpi_use = y_cpi[:nq_cpi]

print(f"  Used for GDP stage-2: {nq_gdp} quarters")
print(f"  Used for CPI stage-2: {nq_cpi} quarters")

# Stage-2 responses
nrep_stage2 = 20000
r2 = stage2irf(y_gdp_use, q1q_gdp, nrep=nrep_stage2, seed=42)
r3 = stage2irf(y_gdp_use, q2q_gdp, nrep=nrep_stage2, seed=42)
r4 = stage2irf(y_gdp_use, q3q_gdp, nrep=nrep_stage2, seed=42)

c2 = stage2irf(y_cpi_use, q1q_cpi, nrep=nrep_stage2, seed=42)
c3 = stage2irf(y_cpi_use, q2q_cpi, nrep=nrep_stage2, seed=42)
c4 = stage2irf(y_cpi_use, q3q_cpi, nrep=nrep_stage2, seed=42)

h2 = np.arange(13)
fig, axes = plt.subplots(3, 2, figsize=(13, 11))

# GDP panels
for ax, rr, ttl, sgn in [
    (axes[0, 0], r2, 'Crude Oil Supply Shock (BVAR)', -1),
    (axes[1, 0], r3, 'Aggregate Demand Shock (BVAR)', 1),
    (axes[2, 0], r4, 'Oil-Market Specific Demand Shock (BVAR)', 1),
]:
    m = sgn * rr['cum']
```

```
    lo68 = sgn * rr['cum_68'][0]
    hi68 = sgn * rr['cum_68'][1]
    lo95 = sgn * rr['cum_95'][0]
    hi95 = sgn * rr['cum_95'][1]
    a68, b68 = np.minimum(lo68, hi68), np.maximum(lo68, hi68)
    a95, b95 = np.minimum(lo95, hi95), np.maximum(lo95, hi95)

    ax.plot(h2, m, 'r-')
    ax.plot(h2, a68, 'b--')
    ax.plot(h2, b68, 'b--')
    ax.plot(h2, a95, 'b:')
    ax.plot(h2, b95, 'b:')
    ax.plot(h2, np.zeros_like(h2), 'k-')
    ax.set_xlim(0, 12)
    ax.set_ylim(-20, 20)
    ax.set_ylabel('Real GDP')
    ax.set_title(ttl)
axes[2, 0].set_xlabel('Quarters')

# CPI panels
for ax, rr, ttl, sgn in [
    (axes[0, 1], c2, 'Crude Oil Supply Shock (BVAR)', -1),
    (axes[1, 1], c3, 'Aggregate Demand Shock (BVAR)', 1),
    (axes[2, 1], c4, 'Oil-Market Specific Demand Shock (BVAR)', 1),
]:
    m = sgn * rr['cum']
    lo68 = sgn * rr['cum_68'][0]
    hi68 = sgn * rr['cum_68'][1]
    lo95 = sgn * rr['cum_95'][0]
    hi95 = sgn * rr['cum_95'][1]
    a68, b68 = np.minimum(lo68, hi68), np.maximum(lo68, hi68)
    a95, b95 = np.minimum(lo95, hi95), np.maximum(lo95, hi95)

    ax.plot(h2, m, 'r-')
    ax.plot(h2, a68, 'b--')
    ax.plot(h2, b68, 'b--')
    ax.plot(h2, a95, 'b:')
    ax.plot(h2, b95, 'b:')
    ax.plot(h2, np.zeros_like(h2), 'k-')
    ax.set_xlim(0, 12)
    ax.set_ylim(-20, 20)
    ax.set_ylabel('CPI')
    ax.set_title(ttl)
axes[2, 1].set_xlabel('Quarters')

plt.tight_layout()
plt.show()
```

```
Stage-2 alignment:
  Quarterly shocks : 203 quarters (q1q, q2q, q3q)
  GDP growth       : 203 quarters
  CPI inflation    : 202 quarters
  First quarter    : 1975 Q1
  Used for GDP stage-2: 203 quarters
  Used for CPI stage-2: 202 quarters
```